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What is Prompt Engineering?
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** Prompt Engineering means writing clear and effective instructions to an AI model so that it gives the best output.

** Prompt Engineering is the skill of asking the right question in the right way to get the best answer from AI.

** AI models like ChatGPT, Gemini, Claude, etc., do not read our mind. They generate output based on the input we give. That input is called a prompt.

Ex:

Poor prompt

Explain java

Good prompt:

Act as java trainer, explain about java programming language to a beginner with simple examples, real-time usecases and give few important interview questions also.

The second prompt gives better output because it clearly says:

Role : JAVA Trainer

Topic : JAVA

Audience : Beginner student

Style : Simple Explanation

Extra Needs : Examples, real-time scenarios, interview questions

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Basic Formula of a Good Prompt
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=> Good prompts usually contains

- 1) Role
- 2) Task
- 3) Context
- 4) Output Format
- 5) Constraints

Ex:

Act as a gen ai trainer
explain RAG and vector databases

for beignners
give step by step explanation,
realtime examples and mini project
use simple english

Role : Tell AI what role it should play while answering your question

Act as a Java developer.
Act as a GEN AI trainer.
Act as a DevOps engineer.
Act as an interviewer.
Act as a career counselor.

Bad Prompt :: Review my code

Good Prompt : Act as a senior Java developer, Review the below Spring Boot code and identify performance issues.

Task :: Clearly mention what you want

Explain
Summarize
Generate
Compare
Debug
Rewrite
Create
Design
Analyze
Convert

Ex :: Explain prompt engineering to beginners with examples.

Context :: Context means background information.

Note: Without context, AI may give a general answer.

Poor prompt : write email

Better Prompt : Write a professional email to students informing them that today GEN AI class is postponed due to a trainer emergency. Mention that the next class schedule will be updated soon.

Output Format :: Tell AI how the answer should look.

Give answer in table format.
Give step-by-step explanation.
Give code with comments.
Give 10 bullet points.
Give JSON response.
Give interview questions and answers.

Constraints : Constraints are rules we give to AI.

Use simple English.

Do not use technical jargon.
Keep it within 10 lines.
Give only Java code.
Do not explain theory.
Use Spring Boot 3.
Use Angular 17.

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Types of Prompts
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- 1) Zero-shot prompting
- 2) One-shot prompting
- 3) Few-shot prompting

In zero-shot prompting, we ask AI directly without giving examples

Ex-1: Explain what is REST API.

Ex-2: Generate 10 interview questions on Spring Boot.

Ex-3: Create poster for GEN AI new batch

one-shot prompting, we give one example and ask AI to follow the same style.

Convert the following sentence into a professional message.

Example:

Input: Class cancelled today.

Output: Dear Students, today's class has been cancelled due to unavoidable reasons. The updated schedule will be shared soon.

Now convert:

Input: Tomorrow demo is there at 8 PM.

Few-Shot Prompting, we give multiple examples.

Ex :

Convert casual messages into professional messages.

Example 1:

Input: Class not there today.

Output: Today's class will not be conducted.

Example 2:

Input: Join fast, demo started.

Output: The demo session has started. Please join as soon as possible.

Now convert:

Input: Fee pending students pay immediately.

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Golden Prompt Template
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Act as a [role].
I want to [task].
Context: [background].
Audience: [target users].
Output format: [table/code/steps/bullets].
Constraints: [simple English/short answer/no jargon/etc.].
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Prompt engineering vs Loop Engineering vs Harness Engineering
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** Prompt Engineering means designing better instructions for an AI model so that we get better results.

PROMPT ENGINEERING = HOW DO I ASK AI CORRECTLY?

** Loop Engineering means designing a repeated cycle where AI can perform a task, take an action, observe the result, and decide what to do next.

LOOP ENGINEERING = HOW CAN AI TRY, CHECK, AND TRY AGAIN?

Suppose you ask: "Find the fastest route to the airport."

A useful system may:

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Get current location
  â†“
Get destination
  â†“
Check available routes
  â†“
Check traffic
  â†“
Compare routes
  â†“
Select best route
  â†“
Give recommendation
```

This is the type of thinking behind loop-based agent systems.

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CODING AGENT EXAMPLE
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User: "Create a Python calculator."

The AI creates code.

Then the system can:

But the student also needs:

- Question paper
- Pen
- Calculator
- Reference material
- Time limit
- Rules
- Invigilator
- Answer sheet
- Evaluation

Note: The brain alone is not enough.

Similarly, an AI agent needs an environment around it.

That environment is the harness.

CUSTOMER SUPPORT AGENT

Customer asks: "Where is my order?"

The AI can be given tools:

```
LLM
|
+-- Search Orders
+-- Check Payment
+-- Check Delivery
+-- Cancel Order
+-- Create Support Ticket
```

Complete system:

```
Customer
  â†“
AI Agent
  â†“
Authentication
  â†“
Understand Question
  â†“
Search Order Database
  â†“
Check Delivery API
  â†“
Get Result
  â†“
Generate Answer
  â†“
Security Check
  â†“
Customer
```

Note : This complete surrounding system is the Harness.

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STAGE 1 â€™ PROMPT ENGINEERING
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Prompt:

You are an HR assistant.

Answer employee questions about company policies.

Use simple English.

Never invent company policies.

If information is unavailable, say:
"I couldn't find this information."

This is Prompt Engineering.

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STAGE 2 â€™ LOOP ENGINEERING
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Employee Question

â€™

AI understands question

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Search HR documents

â€™

Get relevant documents

â€™

AI checks information

â€™

Enough information?

â€™

â€™

No

Yes

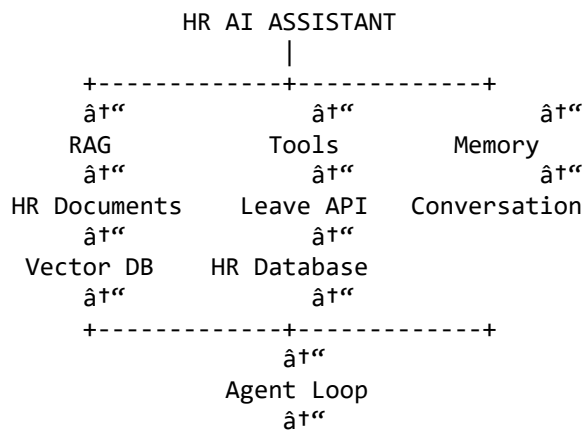
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â€™

Search again Answer

This is Loop Engineering.

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STAGE 3 â€™ HARNESS ENGINEERING
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Guardrails
â†“
Authorization
â†“
Evaluation
â†“
Monitoring
â†“
Answer

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PROMPT ENGINEERING

Main question: "What should I tell the AI?"

Example: "Explain Java to a beginner with simple english."

LOOP ENGINEERING

Main question: "How can AI try, check and retry?"

Example: Generate code â†’ Test â†’ Fix â†’ Test

HARNESS ENGINEERING

Main question: "What environment does AI need to succeed safely?"

Example: Tools + RAG + Memory + Security + Tests

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EASY CAR ANALOGY

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Prompt Engineering:

You tell the driver:
"Go to Hyderabad."

This is the instruction.

Loop Engineering:

Drive
â†“
Check GPS
â†“
Traffic detected
â†“
Change route
â†“
Check GPS
â†“
Continue

Harness Engineering:

Driver

+ Car

+ GPS

+ Road map

+ Traffic information

+ Fuel

+ Brakes

+ Safety systems

+ Rules

+ Sensors